

THE AI TOOLBOX · CLASS 09

Data Visualization (Data Storytelling)

Use charts to **argue**, not just to decorate.

Merit Point Academy · 60 min · press **S** notes · → next ·  to comment

RECAP · WHERE WE LEFT OFF

Last class in 60 seconds

Five enemies, three moves

Missing · normalize · encode — and the type picks the encoding (no fake order!).

Every drop is a claim

Cleaning decisions got reasons; disclosure became habit.

Shortcut learning

The X-ray token: suspiciously good = a hypothesis to attack.

Homework share: the strangest thing you found while cleaning — and your shortcut audit. Your data is now clean and documented... **but it still can't speak.** Today we give it a voice: one conclusion, three honest charts.

Where we're going

- **0–8'** Homework share + four datasets in one disguise
- **8–18'** Charts are arguments · which chart for which job
- **18–28'** Correlation — and its favorite trap
- **28–38'** Honest visuals: the axis that lies · matplotlib's five lines
- **38–46'** Demos: charts in Colab · the AI chart critic
- **46–54'** Two labs: ASCII charts · Anscombe, verified by hand
- **54–60'** Your 3-chart data story, homework & wrap-up
- Quizzes &  comments throughout

By the end: pick the right chart for a claim, compute and distrust correlation appropriately, spot an axis that lies, and tell one clear story with exactly three honest charts.

WARM-UP · ACTIVATE

Quick check · tap an answer

When is a self-driving car's vision least trustworthy — near or far? big objects or small? Where does that answer live?

In the car's user manual

In expert opinion — ask three engineers

Already in your detection log, waiting for the right chart to reveal it

Nowhere; trustworthiness can't be measured

THE STORY

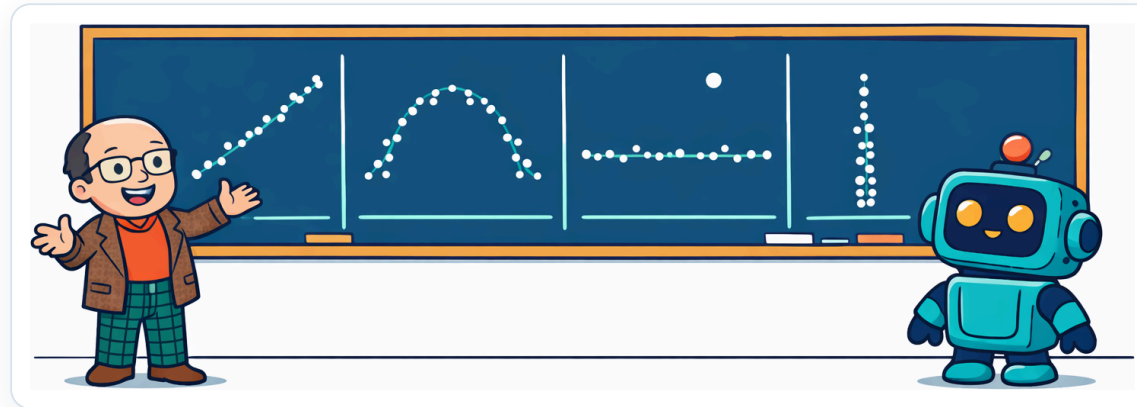
Four datasets, one disguise

In 1973, statistician **Francis Anscombe** built four little datasets with nearly identical statistics — same mean, same variance, same correlation, same best-fit line. By the numbers they were **twins**. Then he plotted them. One was a clean line, one a curve, one had a single wild outlier, one was almost vertical. **The numbers hid everything; only the charts told the truth.**



Speak this story

Neural voice · click to play / pause



You'll meet the actual four on the visual slide — and in Lab 2 you'll **verify his trick yourself**: compute the identical stats, then plot the four shapes. Fifty years later it's still the best two-minute argument for "always plot your data."

CONCEPT 1 · WHAT A CHART IS FOR

A chart is an argument



Histogram

"How is one variable **distributed**?" — where do my detection distances pile up?



Scatter

"How do two variables **relate**?" — does confidence fall as objects shrink?



Correlation

"How **strong** is that relationship?" — one number (r) summarizing the scatter.

KEY IDEA — every chart should answer a question you can state **in one sentence**. If you can't say the sentence, you don't need the chart. (Recognize the discipline? It's Class 2's picture test — now pointed at output instead of input.)

CONCEPT 2 · THE QUESTION PICKS THE CHART

Five jobs, five charts



Trend over time

→ **line chart**. Sales by month, temperature by hour.



Compare categories

→ **bar chart**. Detections per object class.



Parts of a whole

→ **pie** (sparingly!). Market shares — only when parts sum to 100%.



Distribution

→ **histogram / box plot**. Ages, distances, scores.



Relationship

→ **scatter**. Confidence vs. size; study hours vs. grades.



The rule

State the question first — **the question picks the chart**. (Sound familiar? "The type picks the tools" — Class 7.)

CHECK · PICK THE CHART

Quick check · tap an answer

Your claim: 'the detector's confidence drops as objects get smaller.' Which chart carries that argument?

A pie chart of object classes

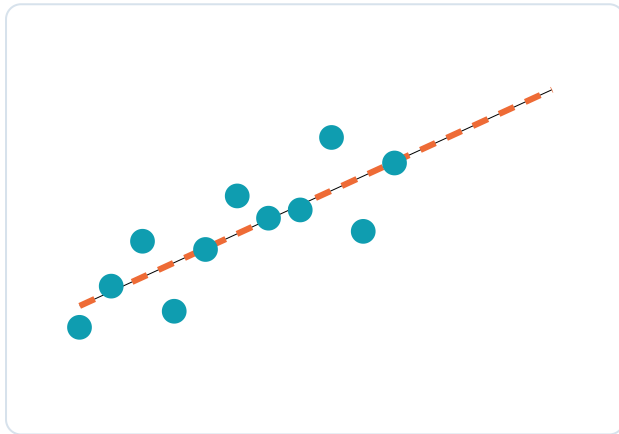
A scatter plot: box size on x, confidence on y

A bar chart of average confidence

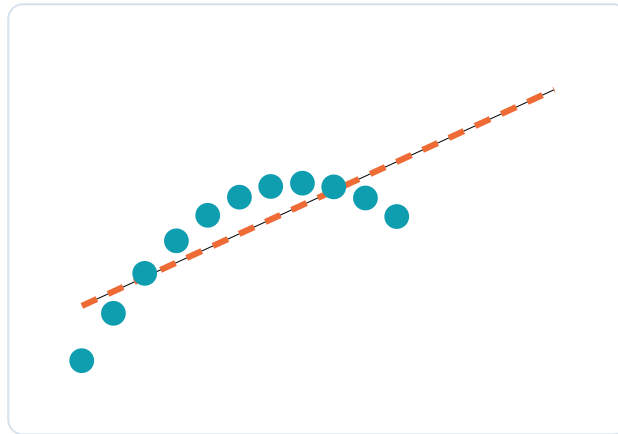
A line chart of confidence over time

Same stats, four different truths

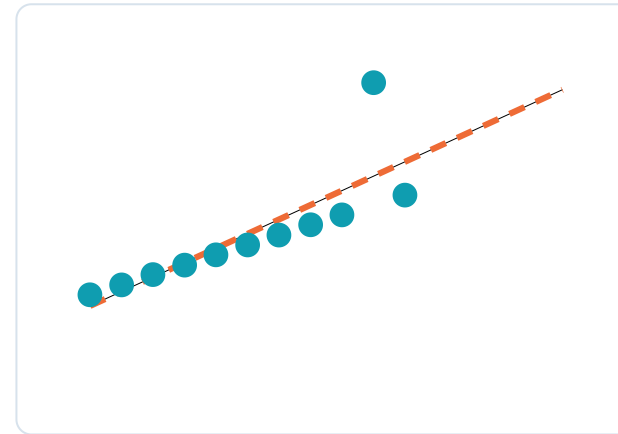
These are Anscombe's actual numbers. All four: $\text{mean}(x)=9$, $\text{mean}(y)=7.5$, correlation $r \approx 0.816$, best-fit line $y = 3 + 0.5x$ (dashed):



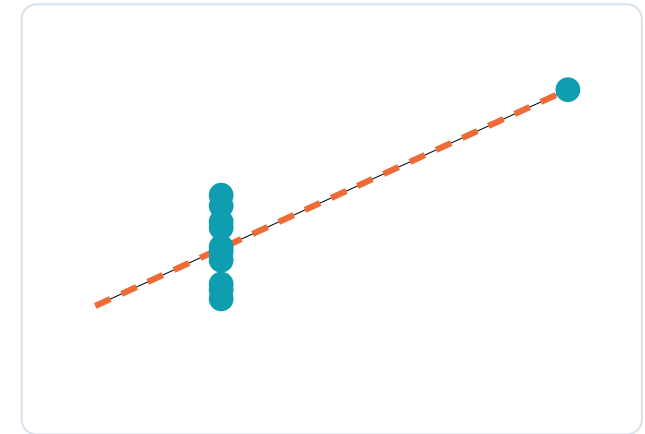
I · linear



II · curved



III · outlier



IV · vertical

Takeaway: identical summary numbers can hide totally different shapes — a curve, a killer outlier, a vertical stack. **Always plot your data** before you trust any statistic about it. `describe()` is the beginning of looking, never the end.

CONCEPT 3 · THE STRENGTH NUMBER

Correlation: one number for a relationship

$$r \rightarrow +1$$

Rise together, tightly. Study hours vs. grades (usually!).

$$r \rightarrow 0$$

No *linear* pattern. Shoe size vs. music taste.

$$r \rightarrow -1$$

One rises, the other falls. Object size vs. distance in our log.

! Trap 1 · correlation $\neq$ causation

Ice-cream sales correlate with drownings — summer causes both. An r , however large, never proves one thing *drives* the other. (Class 5's Titanic quiz, formalized.)

! Trap 2 · r only sees straight lines

Anscombe's panel II: a perfect curve, $r = 0.816$ — same as the line! Strong nonlinear patterns can hide behind any r . The fix is always the same: **plot it**.

Quick check · tap an answer

Cities with more firefighters have more fire damage (strong positive r). Therefore firefighters cause damage. What went wrong?

Nothing — the data has spoken

The r must have been computed incorrectly

A lurking variable: bigger fires bring both more firefighters AND more damage — correlation isn't causation

Firefighters should be reduced to test it

The axis that lies

Same three detectors, same numbers — 88%, 90%, 92% accuracy. Watch the y-axis do the lying:

y from 0



A

B

C

Honest: axis starts at 0
the three detectors are nearly tied

y from 87



A

B

C

Misleading: axis starts at 87
C looks 3× better — same numbers!

The honest-chart checklist: axes start at sensible values (bars: usually 0) · units labeled · every point's meaning stated · **caption self-contained** — a reader should get the message without you narrating. Break these to make a point *look* bigger, and you've joined the X-ray token on the dark side.

THE TOOL · FIVE LINES TO A REAL CHART

matplotlib, gently

Histogram

```
import matplotlib.pyplot as plt
plt.hist(df["dist_m"], bins=20)
plt.title("Detection distances")
plt.xlabel("distance (m)")
plt.ylabel("count"); plt.show()
```

Scatter

```
plt.scatter(df["box_area"],
            df["conf"], alpha=0.5)
plt.title("Confidence vs. size")
plt.xlabel("box area (px2)")
plt.ylabel("confidence"); plt.show()
```

Notice what's NOT optional: three of the five lines are title and axis labels — the same rule since your first Colab chart (Class 5). The plotting is one line; the **honesty** is the other four. Correlation, when you need the number: `df["a"].corr(df["b"])`.

When is perception least confident?

1

Histogram

Distances pile up at 5–20 m — but a long tail stretches past 60 m. *"Far detections exist and matter."*

2

Scatter

Confidence vs. box size: a clear upward drift — small boxes, shaky confidence. *"Size drives certainty."*

3

Correlation

$r(\text{size, conf}) \approx +0.65$ · $r(\text{distance, conf}) \approx -0.79$ — the numbers behind the pictures (computed from Lab 1's 20 rows: verify them in Lab 1 Challenge 2).

Conclusion ▶ Our detector is least trustworthy exactly where stakes are high: **small, far objects** — which is where a night-time pedestrian first appears.

Anatomy of a data story: each chart answers one sentence; together they corner a single conclusion — and that conclusion is Class 2's strong question, **now with evidence**. This is the story-shape your project copies today.

DISCUSS · 5 MINUTES · NUMBERS THAT HIDE

A great summary number hiding a problem

Anscombe's mean hid four different shapes. What's a number from **your** world that could look great in a summary while hiding trouble underneath?

Find one

A class average hiding a struggling half? An app's average rating hiding a review-bomb? A model's 95% accuracy hiding one always-missed class?

Reveal it

Which chart exposes the hiding place? And — have *you* ever been fooled by a chart? What was the trick?

Pairs, 3 minutes — best hidden-number + revealing chart in .

LIVE DEMO 1

The data story, built live in Colab

From cleaned log to three captioned charts in ten minutes — the exact flow of tonight's deliverable.

```
You ▶ df["box_area"] = df.box_w * df.box_h # feature engineering, Class 8!
```

```
You ▶ plt.hist(df["dist_m"], bins=20) + title + labels → chart 1, captioned
```

```
You ▶ plt.scatter(df["box_area"], df["conf"], alpha=0.5) → chart 2, captioned
```

```
You ▶ df[["box_area", "dist_m", "conf"]].corr().round(2) → the numbers for chart 3
```

```
You ▶ plt.savefig("story1.png", dpi=150) – export for your poster/README
```



colab.research.google.com

LIVE DEMO 2

The AI chart critic

Class 4's mentor hat, aimed at your charts: generate faster, then get audited for honesty.

You ▶ Write matplotlib code for a histogram of `dist_m` with proper title, axis labels, and units.

You ▶ Here's my chart code + caption: [paste]. Critique it against this checklist: honest axes, labeled units, self-contained caption.

You ▶ My claim is 'small objects get low confidence.' Which chart best supports it – and what chart would OVERSTATE it?

You ▶ Act as a skeptical reviewer: how could this chart mislead someone? Fix it.



chatgpt.com

Build a histogram and a scatter — out of text

No library, no magic: a histogram is **counting into buckets**; a scatter is **placing points on a grid**. Build both from raw characters and the concepts become yours forever.

```
1 # The detection log's distances (m) and confidences - cleaned, Class 8 style
2 dist = [8.2, 15.0, 14.1, 22.4, 9.5, 13.0, 7.8, 16.2, 21.0, 6.9,
3         31.5, 44.0, 12.3, 55.8, 18.7, 26.4, 38.2, 9.9, 48.5, 11.2]
4 conf = [0.94, 0.88, 0.91, 0.79, 0.62, 0.95, 0.58, 0.90, 0.83, 0.97,
5         0.71, 0.55, 0.92, 0.42, 0.85, 0.74, 0.61, 0.96, 0.47, 0.93]
6
7 # — HISTOGRAM = count into buckets —————
8 print("Distance distribution (each # = 1 detection)")
9 for lo in range(0, 60, 10):
10     n = sum(lo <= d < lo + 10 for d in dist)
11     print(f"{lo:2d}-{lo+10:<3d}m | {'#' * n}")
12
```



Run



Explain each line



Explain



Debug



Improve



Mira

Python loads on first Run

► Run the code to see the output here.

Anscombe's quartet, verified by hand

Don't take the story on faith. Compute the stats yourself — means, correlation, fit line — then plot the four shapes. **The numbers say twins; your eyes will disagree.**

```
1 # Anscombe's actual 1973 numbers - all four sets
2 x123 = [10, 8, 13, 9, 11, 14, 6, 4, 12, 7, 5]
3 sets = {
4     "I ": (x123, [8.04, 6.95, 7.58, 8.81, 8.33, 9.96, 7.24, 4.26, 10.84, 4.82, 5.68]),
5     "II ": (x123, [9.14, 8.14, 8.74, 8.77, 9.26, 8.10, 6.13, 3.10, 9.13, 7.26, 4.74]),
6     "III": (x123, [7.46, 6.77, 12.74, 7.11, 7.81, 8.84, 6.08, 5.39, 8.15, 6.42, 5.73]),
7     "IV ": ([8, 8, 8, 8, 8, 8, 8, 19, 8, 8, 8],
8             [6.58, 5.76, 7.71, 8.84, 8.47, 7.04, 5.25, 12.50, 5.56, 7.91, 6.89]),
9 }
10
11 def mean(v): return sum(v) / len(v)
12 def corr(xs, ys): # Pearson's r from scratch
```



Run



Explain each line



Explain



Debug



Improve



Mira

Python loads on first Run

► Run the code to see the output here.

YOUR PROJECT

A 3-chart data story

Use **exactly three charts** to make **one clear, defensible conclusion** about your own dataset.

- State your conclusion in **one sentence** — first, before any plotting
- Pick the 3 charts that genuinely support it (the question picks the chart!)
- Build them with titles, labeled axes, units
- Write a **self-contained caption** for each

Deliverable

A "data story" poster: 3 captioned charts + the one-sentence conclusion.

Success looks like

The 3 charts genuinely support the one claim — and axes and captions are honest and clear.

BUILD IT NOW · HANDS-ON

Conclusion first, charts second

1. Open your Class-8 Colab (cleaned data ready) → write your **one-sentence conclusion** as a markdown cell
2. Chart 1 — usually a **histogram**: show the landscape your claim lives in
3. Chart 2 — usually a **scatter**: show the relationship doing the work
4. Chart 3 — the **strength or comparison**: `.corr()`, or a grouped bar
5. Caption each in one sentence: *what it shows* → *why that supports the claim*
6. The honesty pass: axes, units, labels — then **Run all**, savefig, commit to GitHub

Colab

matplotlib

ChatGPT / Claude (chart critic)

GitHub

The cut rule: if a chart doesn't pull its weight for the claim, cut it — **three strong beats five weak**. (And a chart you can't caption in one sentence is telling you something.)

What a finished data story looks like

Conclusion: *"Our detector is least reliable on small, distant objects — exactly where night-time pedestrians first appear."*



Chart 1 · histogram

Distances, 20 bins, x-labeled in metres.

Caption: "30% of detections are beyond 25 m — the far zone isn't rare, so failures there matter."



Chart 2 · scatter

Box area vs. confidence, $\alpha=0.5$, both axes united. **Caption:** "Confidence sinks with size — below $\sim 2,000$ px², most detections fall under 0.7."



Chart 3 · correlation

r table: size $\leftrightarrow$ conf +0.65, distance $\leftrightarrow$ conf -0.79.

Caption: "The drift is strong, not a squint: size and distance together explain most confidence loss."

The bar: each caption is *what it shows* → *why it supports the claim*, the three charts hit distribution → relationship → strength, and nothing needs the author standing next to it. This poster is slide 3 of your final presentation (Class 15), finished seven weeks early.

Today's vocabulary

Histogram — counts in buckets: the shape of one variable's distribution. Bin width is a judgment call.

Correlation (r) — linear-relationship strength, $-1 \dots +1$. Blind to curves; silent on causes.

Truncated axis — a y-axis that skips zero to inflate differences: the most common chart lie.

Scatter plot — two variables, one point per row: the shape of a relationship.

Lurking variable — the hidden driver behind a correlation (summer → ice cream + drownings).

Data story — one conclusion, a few honest charts, self-contained captions. Your deliverable — and Class 15's opening act.

RECOMMENDED HOMEWORK

Before next class

1. **Finish your 3-chart data story** (conclusion + captioned charts, honesty pass done) — submit the Colab link in Google Classroom, commit the PNGs to GitHub.
2. **Chart-lie hunt:** find one misleading chart in the wild (news, ads, social media). One sentence: what's the trick, and what's the one-line fix?
3. **Re-run Lab 2's challenge:** remove set III's outlier and report what happens to r — one sentence on what that teaches about outliers and correlation.

★ Stretch (optional)

Meet the **Datasaurus Dozen** — Anscombe's modern heir: thirteen datasets with identical stats, one of which is a *dinosaur*. Scroll the animation and grin.

[autodesk research · Datasaurus](#)

The chart-lie hunt is the habit that outlives the course — once you see truncated axes, you can't unsee them. Questions →  on any slide.

TODAY'S LINE TO REMEMBER

"The greatest value of a picture is when it forces us to notice what we never expected to see."

— John Tukey, father of exploratory data analysis

Anscombe proved it with four plots. Tonight, your three charts might surprise **you** first.



Speak this

WRAP-UP

Three things to keep

- Charts make **arguments** — every chart answers a one-sentence question
- **Always plot your data** — identical stats can hide four different truths
- Honest axes, labeled units, self-contained captions — **or it's decoration**

Exit ticket: the one-sentence conclusion of your data story — post it in .

Next · Class 10 — Stage 4 begins: your first model that **predicts**. The data stage is complete.

BACKUP MATERIAL

Extra slides — if there's time

Deeper dives and extra practice. Skip in a tight hour; pull up on demand or for a longer session.

The liar's gallery

three classic chart tricks

The extended chart menu

box plots, lines, heatmaps

Bonus quiz

one more check

Three classic chart tricks (and the catch)



The truncated axis

y starts at 87, differences triple visually. **Catch:** read the axis numbers *first*, always.



The cherry-picked window

"Sales up 40%!" — chart starts at last year's crash. **Catch:** ask what happens if the x-axis extends left.



The double y-axis

Two unrelated scales overlaid until they "correlate." **Catch:** two y-axes = two stories; demand them separately.

The general antidote: a chart is a claim (concept 1) — so apply Class 4's discipline to it: *who made this, what's the source, what would the honest version look like?* Verification isn't just for text.

Three more charts worth knowing



Box plot

`describe()` as a picture: median, quartiles, and outliers flagged automatically. Best for comparing distributions across groups — `df.boxplot(column="conf", by="object")`.



Line chart

The trend specialist — but only when x is *ordered* (time, distance). Lines on unordered categories draw connections that don't exist.



Heatmap

A whole correlation table at a glance — `plt.imshow(df.corr())`. The instant map of which variables move together.

Still the same rule: **the question picks the chart**. New chart types are new answers, not new decorations — add one to your story only if it carries a sentence the core three can't.

Quick check · tap an answer

A news chart shows crime 'exploding' — a steep red line. You check: the y-axis runs 41.0 to 42.5 incidents, and the x-axis covers exactly the two worst months of a flat year. Diagnose the tricks.

No tricks — steep lines mean big changes

One trick: bad colors

Two stacked tricks: truncated axis magnifying a ~3% wiggle, plus a cherry-picked time window

The data must be fake